Chapter 1: Introduction to Robotics

***What is a Robot?

A **robot** is a programmable, intelligent machine capable of **sensing**, **planning**, and **acting** to interact with and affect its environment. It may be **autonomous**, **semi-autonomous**, or **remotely controlled**, depending on its level of decision-making capability and control.

***Is a programmable device always a robot?

A robot is a specific type of programmable device that:

1. **Can sense its environment** (using sensors)
2. **Processes information** (using a controller or processor)
3. **Acts or moves physically** in the real world (motors, arms, wheels, etc.)
4. Is usually **autonomous or semi-autonomous** (can work without human control)

Examples of programmable devices that are **not robots**:

Device	Why it's not a robot

Washing Machine	It doesn't sense or adapt to environment in a complex way
TV Remote	No sensors, no physical actions
Microwave Oven	Follows pre-programmed steps, no autonomy or mobility
Arduino Board Alone	It’s a controller - not a robot until connected with actuators/sensors
Calculator	Just processes data, no interaction with physical world

Examples of programmable devices that **are robots**:

Device	**Why it is a robot**
Robot vacuum (e.g., Roomba)	Senses environment, makes decisions, moves
Industrial arm robot	Executes tasks with physical actions (welding, painting)
Humanoid robot (e.g., Pepper, ASIMO)	Interacts, moves, and processes sensory input
Line-following robot	Follows a path using sensors and motors

In short:

- **All robots are programmable devices**,
- But **not all programmable devices are robots**.

Objective of Robotics

🔹 1. Industry Automation

Definition:
Using robots to automate industrial processes such as manufacturing, assembly, welding, painting, packaging, etc.

Purpose:

- Increase productivity
- Improve precision and quality
- Operate in hazardous or repetitive environments
- Reduce human labor cost

Example:
Articulated robot arms in car manufacturing plants (welding, bolting, painting).

🔹 2. Smart, IoT and Embedded Systems

Definition:
Robots integrated with **IoT (Internet of Things)** and **smart embedded systems** for real-time sensing, communication, and decision-making.

Purpose:

- Enable robots to collect and share data
- Support cloud-based control and updates
- Adapt to environments dynamically
- Connect multiple systems for collaborative tasks

Example:

- A **smart home cleaning robot** that maps rooms using sensors and updates cleaning schedules via cloud.
- **IoT-connected agriculture robots** that monitor soil conditions and apply water/fertilizer intelligently.

🔹 3. Efficient Task-Specific Robots

Definition:
Robots designed for **specific roles** to optimize performance, reliability, and energy efficiency.

Purpose:

- Maximize task efficiency
- Reduce design complexity and cost
- Focused skill development for particular domains (e.g., surgery, mining)

Example:

- **Surgical robots** like the Da Vinci system—precise, task-specific for minimally invasive surgery.
- **Delivery drones**—optimized for short-range parcel delivery with GPS-based planning.

Why Intelligent Machines?

🔹 1. Energy Saving

Why it matters:
Traditional machines often operate at full power regardless of the situation. Intelligent machines **optimize energy usage** based on real-time conditions.

Example:
A robot vacuum only cleaning dirty spots after identifying them with sensors saves both time and battery.

🔹 2. Accuracy

Why it matters:
Precision is critical in fields like surgery, electronics manufacturing, and aerospace.

Example:
Surgical robots like Da Vinci systems perform microscale operations with sub-millimeter accuracy.

🔹 3. Convenience

Why it matters:
Robots take over mundane tasks, freeing humans for creative or strategic roles.

Example:
Home robots like Alexa-controlled vacuums or eldercare assistive robots simplify daily living.

🔹 4. Efficiency

Why it matters:
More output in less time, with fewer resources.

Example:
Automated warehouses use AGVs (automated guided vehicles) for real-time item delivery- far faster than human workers.

🔹 5. Adaptability in Dynamic Environments

Why it matters:
Real-world conditions are unpredictable. Intelligent machines **sense and respond** to changes on the fly.

Example:
A Mars rover rerouting around rocks or terrain changes based on its own sensory feedback.

🔹 6. Perform Dull, Dirty, Difficult, Dangerous (4D) Jobs

Why it matters:
Protects humans from harm or boredom and extends operation into hostile environments.

Examples:

- **Dull:** Repetitive factory inspection
- **Dirty:** Sewer inspection robots
- **Difficult:** Climbing robots for skyscraper maintenance
- **Dangerous:** Bomb disposal or mine clearance robots

***What is 4D in robotics and how can robotics be useful in handling this term?

The 4D (Dull, Dirty, Difficult, and Dangerous) concept refers to tasks that are tedious, hazardous, or physically demanding, making them unsuitable or even impossible for humans to perform. Robotics technology is ideally suited for performing such tasks, as it allows for automation and remote control of equipment, thus reducing the risk to human operators. Here's how each of the 4D categories can be applied to robotics:

Dull: Tasks that are repetitive and monotonous, such as assembly line work, can be automated using robots. This reduces the risk of errors and allows humans to focus on more creative and strategic tasks.

Dirty: Tasks that involve exposure to hazardous materials, such as cleaning up radioactive waste or handling toxic chemicals, can be performed by robots. This protects human workers from exposure to dangerous substances and reduces the risk of contamination.

Difficult: Tasks that require physical strength or agility, such as lifting heavy objects or working in tight spaces, can be performed by robots. This reduces the risk of injury to human workers and increases efficiency by allowing robots to perform tasks that would be difficult or impossible for humans.

Dangerous: Tasks that are hazardous to human life, such as working in mines, handling explosives, or conducting search and rescue operations in disaster zones, can be performed by robots. This reduces the risk of injury or death to human workers and allows robots to perform tasks that would be too dangerous for humans to attempt.

Overall, the 4D concept highlights the value of robotics in automating tasks that are dull, dirty, difficult, or dangerous, thus improving safety, efficiency, and productivity in various industries.

***Laws of AI Robots

🔹 **1. A robot must not harm a human being, nor through inaction allow one to come to harm.**

Meaning:
Robots should **never hurt humans**. Even if the robot does nothing and a human gets hurt, that's also wrong.

Example:
If a robot sees a person about to fall into a hole, it **must warn or save** them. It can't just stand there doing nothing.

🔹 **2. A robot must always obey humans, unless that goes against the first law.**

Meaning:
Robots should **listen to what humans say**, but only if it **doesn't hurt another human**.

Example:
If someone tells a robot, "Punch that person," it should **refuse** because that would hurt a human and break Law 1.

🔹 **3. A robot must protect itself from harm, unless that goes against Laws 1 or 2.**

Meaning:
Robots should **protect themselves**, but **humans come first**.

Example:
If a robot is in a fire, it should escape **only if no humans are in danger**. If a person is inside, it must try to help even if it might get damaged.

🔹 **4. A robot should always have a kill switch.**

Meaning:
There should always be a **way to turn the robot off**, in case it goes out of control.

Example:
If the robot malfunctions or becomes dangerous, a **human can press the kill switch** to stop it immediately.

📌 Why These Laws Matter

These laws are like a **safety rulebook** for robots.
They make sure robots:

- **Don’t hurt us**
- **Listen to us**
- **Stay safe** (but not more important than us)
- **Can be stopped if needed**

***Thumb Rules on the decision of a Robot Uses

🔹 The first rule to consider, what is known as the **4D of Robotics**, i.e. is the task dirty, dull, dangerous, or difficult? If so, a human will probably not be able to do the job efficiently. Therefore, the job is appropriate for automation or for robotic labor.

🔹 The second rule is that a robot may **not leave a human jobless**. Robotics and automation must serve to make our lives more enjoyable, not miserable.

🔹 A third rule involves **asking whether you can find people who are willing to do the job**. If not, the job is a candidate for automation and Robotics.

🔹 A fourth rule of thumb is that the use of robots or automation must **make short-term and long-term economic sense**.

***Uncrewed Vehicle (Learn Full Forms)

🔹 **1. Remote Control Vehicle (RC)**

A vehicle controlled **manually by a human** using a remote.

Example: RC cars, RC helicopters, RC planes

🔹 **2. Unmanned Ground Vehicle (UGV)**

A robot that **moves on the ground** without a person inside.

Example: Bomb disposal robots, Military ground robots, Surveillance ground robots, Rescue robots, **BRACU Mongol Tori**

🔹 **3. Unmanned Aerial Vehicle (UAV)** ***(aka Drone)***

A flying robot controlled remotely or automatically.
Example: Surveillance drones, Mapping drones

🔹 **4. Unmanned Combat Aerial Vehicle (UCAV)**

A **military drone** that can **attack targets** without a pilot onboard.
Example: Missiles, drones used in airstrikes or defense missions like Kamikaze drone, Loitering Munition

🔹 **5. Small/Miniature UAV (SUAV)**

A **small-sized drone**, usually used for short-range, lightweight tasks.
Example: Pocket-sized drones used for indoor surveillance or hobby flying like DJI Mavic

🔹 **6. Delivery Drone**

A drone designed to **carry and deliver items** to customers.
Example: Amazon's delivery drones dropping packages at your doorstep.

🔹 **7. Micro Air Vehicle (MAV)**

A **tiny flying robot**, often the size of a bird or insect.
Example: Used in spying or indoor exploration where space is tight.

🔹 **8. Target Drone**

A drone used for **military training** to practice targeting or shooting.
Example: Flying targets in air force shooting practice.

🔹 **9. Autonomous Spaceport Drone Ship**

A **robotic ship** used for **landing rockets in the ocean**.
Example: SpaceX rocket boosters land on these after missions.

🔹 **10. Unmanned Surface Vehicle (USV)**

A **robotic boat** that moves on water **without a crew**.
Example: Used for ocean data collection or naval patrol.

🔹 **11. Unmanned Underwater Vehicle (UUV)**

A robot that moves **underwater without people**.
Example: Used in ocean research or submarine surveillance.

🔹 **12. Remotely Operated Underwater Vehicle (ROUV)**

A **wired underwater robot**, controlled by an operator from above.
Example: Used to inspect underwater pipelines or shipwrecks.

- **13. Autonomous Underwater Vehicle (AUV)**

An underwater robot that can **navigate and work by itself-** no remote needed.
Example: Mapping the ocean floor or monitoring pollution, **BRACU Duburi**.

- **14. Uncrewed Spacecraft (Space Probe or Robotic Spacecraft)**

A **robotic spacecraft** sent to explore space **without astronauts**.
Example: NASA's Voyager or Mars rovers.

How Uncrewed Vehicles Are Useful

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1. Do Dangerous Jobs

Uncrewed vehicles can go where it's **too risky** for humans.

- They can enter war zones, deep oceans, burning buildings, or space.

Example: Bomb disposal robots, Mars rovers, underwater pipeline inspection.

2. Save Human Lives

They take over risky tasks in **military**, **rescue missions**, or **disaster zones**.
Example: Drones help find people after earthquakes or floods.

3. Faster and More Efficient

UVs can work **24/7** without breaks and often **faster than humans**.
Example: Delivery drones can bring packages within minutes.

4. Reach Places Humans Can’t Go

They go **deep underwater, high in the sky**, or even to **outer space**.
Example: Space probes exploring Mars, underwater robots studying sea creatures.

5. Collect Data and Monitor

UVs are used for **surveying, mapping, weather forecasting**, and **military spying**.
Example: Aerial drones take pictures of forests, farms, or enemy locations.

6. Reduce Cost

One robot or drone can replace **many humans**, saving money in the long term.
Example: Autonomous tractors reduce the need for large farm crews.

7. Support Smart Industries

UVs are part of **Industry 4.0**, helping factories become smarter and more automated.
Example: Ground robots move goods inside warehouses like Amazon.

8. Useful in War and Defense

The military uses UCAVs and surveillance drones to **protect borders** and **attack enemies** safely.
Example: Drones flying silently to monitor enemy movements.

Parallel Robot

A **Parallel Robot** is a robot where the end-effector (the part that does the work) is **connected to the base by multiple arms** working together, usually in parallel. Unlike serial robots (like robot arms with joints one after another), **parallel robots are stronger, faster, and more precise**- but have a smaller working area.

🔹 **1. Flight Simulator**

This is a **parallel robot** that **moves in 6 degrees of freedom** (forward-backward, up-down, side-to-side, and rotations).
Usually built using a **Stewart Platform** (a common type of parallel mechanism with 6 legs).

Why Parallel?
Because it needs to move like a real aircraft-**very fast and very accurately**-to.

🔹 **2. Milling Machine (Parallel Kinematic Machine Tool)**

This is a **precision machine** used to **cut materials** like metal or plastic.
Some modern milling machines use **parallel kinematics** for **better stiffness, speed, and accuracy** than traditional machines.

Why Parallel?
Because it gives **stable and precise movement**, perfect for shaping hard materials.

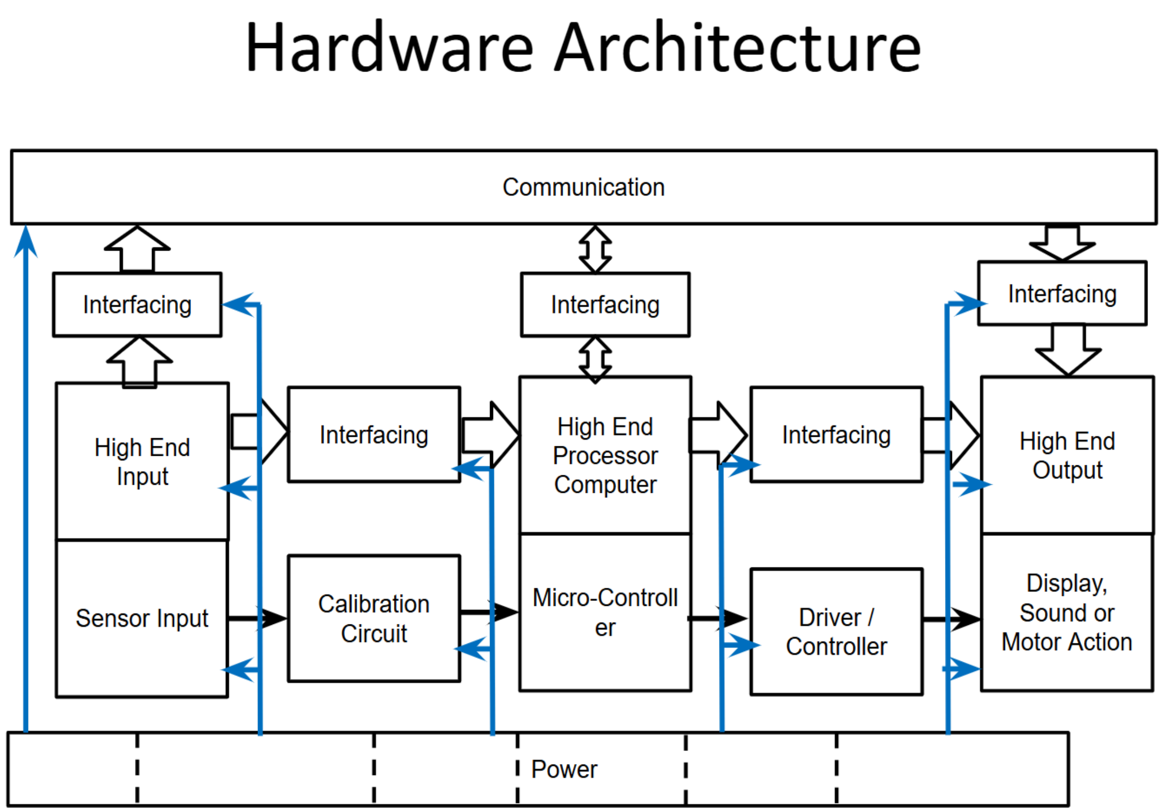
🔹 **3. Cable-Driven Parallel Robot**

This robot uses **tensioned cables instead of solid arms** to move a platform in space (like a spider web).
It's like a **puppet controlled by strings**, but smarter and motor-controlled.

Why Parallel?
Because **multiple cables pull in coordination**-giving high-speed, flexible motion over a **large area**.

📌 **Where it's used?**

- Moving cameras in sports stadiums
- Large-scale 3D printing
- Warehouse picking system



This diagram represents a robot hardware architecture, showing how various components-inputs, processing units, and outputs-interact through interfacing, communication, and power. Here's a detailed explanation of each block:

Bottom Layer: Power

- This supplies electrical energy to all components: sensors, processor, controllers, and actuators.
- Power distribution is crucial for stable real-time operation of all sensors and actuators.

Left Side: Input System

- **Sensor Input**

- Raw data acquisition from the environment (e.g., temperature, proximity, acceleration).
- Where sensor data plays a key role in robotic perception.

- **Calibration Circuit**

- Conditions and calibrates raw sensor signals (e.g., filtering, scaling, bias correction).
- Calibration ensures accurate measurements.

- **High-End Input**

- Advanced sensor data (e.g., from LIDAR, stereo cameras, depth sensors).
- These may require preprocessing or dedicated hardware to extract meaningful features.

- **Interfacing (Input Side)**

- Converts and transmits sensor/calibration output into a format readable by processing units (e.g., ADCs, multiplexers).
- This interface handles electrical compatibility, protocols, and synchronization (e.g., UART, I2C, SPI).
- Covered in Springer Handbook and Peter Corke for sensor-processor interfacing techniques.

Center: Processing System

🔹 **High-End Processor / Computer**

- The main computing unit (e.g., CPU, GPU, embedded PC) handles high-level processing: perception, planning, decision-making.
- Handles data fusion, SLAM, path planning, AI tasks.

🔹 **Microcontroller**

- Real-time, low-latency control logic, such as issuing PWM signals or managing motor states.
- Works closely with the driver circuit.

🔹 **Interfacing (Processing Side)**

- Manages data exchange between processor, memory, controllers, and interfaces with communication buses.

- Necessary for synchronization and protocol translation.

Right Side: Output System

◆ **Driver / Controller**

- Amplifies control signals from the microcontroller to drive actuators (motors, servos, speakers).
- Includes power electronics (H-bridges, MOSFET drivers).

◆ **High-End Output**

- High-power or complex output devices (e.g., robotic arms, robotic hands, haptic displays, multi-DOF motors).
- Might include hardware abstraction or feedback systems.

◆ **Display, Sound, or Motor Action**

- Final output medium, providing visible, audible, or mechanical actions (LCD, buzzer, wheels).
- This is the “action” layer of the sense-plan-act model from Murphy.

◆ **Interfacing (Output Side)**

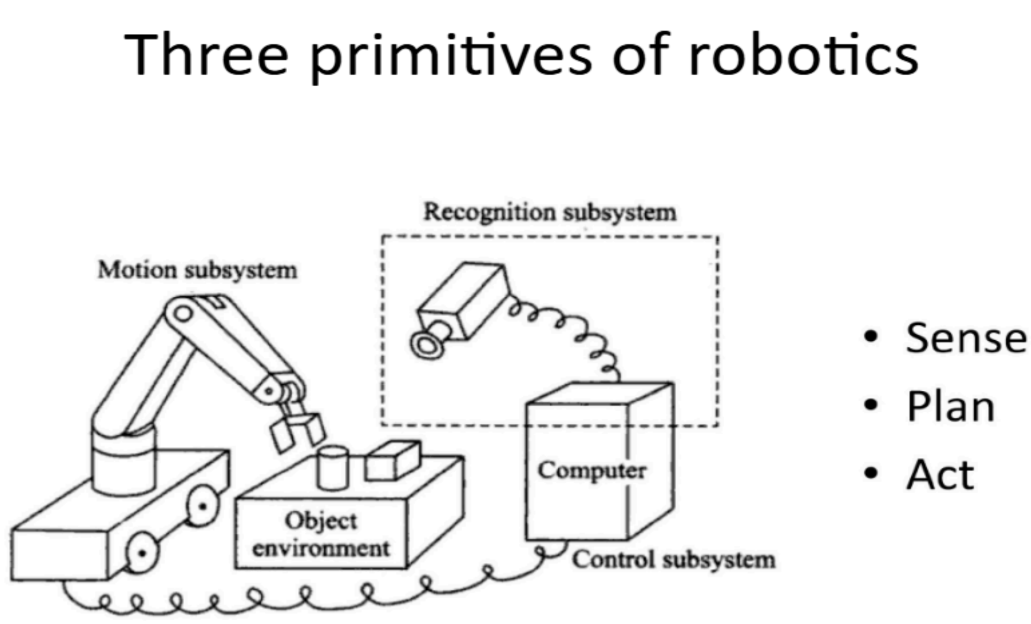
- Converts processor/control signals into compatible electrical form for actuators.
- Helps prevent overloads and signal mismatches.

Top Layer: Communication

- Connects interfacing units and main processor for synchronized operation.
- Can be serial, parallel, bus-based (CAN, RS-485), or wireless (Wi-Fi, Bluetooth).

🔁 **Summary of Flow:**

Sensors → Calibration → Interfacing → Microcontroller & Processor → Control Signals → Driver → Actuators/Output



***Robotic Paradigms

The robotic paradigms- Hierarchical (Deliberative), Reactive, and Hybrid (Deliberative/Reactive)- are foundational frameworks for robot control and decision-making. Each paradigm represents a different approach to how robots perceive, plan, and act.

1. Hierarchical / Deliberative Paradigm

Core Idea:

"Sense → Plan → Act”

This paradigm involves a robot that uses a model of the world to plan its actions based on a specific goal. The robot processes sensory input, generates a plan, and then executes the plan to achieve the desired outcome. This approach is often used in industrial applications, where robots are programmed to perform tasks in a structured environment.

Characteristics:

- Centralized processing and decision-making.
- The world model is maintained (global map).
- Planning is **slow but thorough**.
- Execution happens **after** full deliberation.
- Suitable for **structured** environments.

Example:

- Path planning in a known maze using A*.
- A warehouse robot that plans a full route before moving.

Robot Primitives	INPUT	OUTPUT
SENSE	Sensor Data	Sensed Information
PLAN	Information [sensed and/or cognitive]	Directives
ACT	Directives	Actuator Commands

2. Reactive Paradigm

Core Idea:

"Sense → Act"

This paradigm involves a robot that responds to its environment in real-time without building an explicit model of the world. The robot reacts to sensory input by selecting an appropriate behavior or action from a predefined set of options. This approach is often used in mobile robotics and autonomous vehicles, where the robot needs to navigate a complex and dynamic environment.

Characteristics:

- No internal map or central planner.
- Consists of **behaviors** triggered by perceptions (subsumption architecture).
- **Fast and robust**, works well in dynamic environments.
- Inspired by biological systems (insects, animals).

Example:

- A robot that avoids obstacles and follows walls without knowing the whole layout.

- Roomba vacuum: reacts to bumps and changes direction.
- Automatic Motion-Detecting Doors

Robot Primitives	INPUT	OUTPUT
SENSE	Sensor Data	Sensed Information
PLAN		
ACT	Sensed Information	Actuator Commands

3. Hybrid Deliberative/Reactive Paradigm

Core Idea:

Combines "Sense → Plan → Act" and "Sense → Act"

This paradigm combines elements of both the deliberative and reactive paradigms. The robot builds a model of the world and uses it to plan its actions, but also responds to sensory input in real-time to adjust its plan as necessary. This approach is often used in complex environments where the robot needs to be able to adapt to changing conditions, such as search and rescue missions or space exploration.

Characteristics:

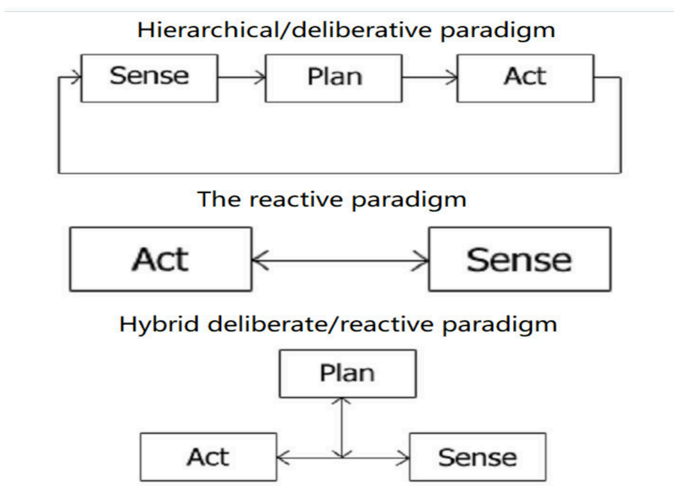
- Maintains **partial world model**.

- Top layer does **goal planning**, bottom layer handles **real-time responses**.

- Balances flexibility (reactivity) and intelligence (deliberation).
- Modular design: planner, controller, behavior executor.

Example:

- A delivery robot plans a path to a room but reacts to obstacles or people in real time.
- Mars rovers: global goal planning, but obstacle avoidance is reactive.



Each of the paradigms has its strengths and weaknesses, and the choice of which paradigm to use depends on the specific application and the requirements of the robot. By understanding these different paradigms, researchers and engineers can design robots that are better suited to perform specific tasks and operate effectively in different environments.

***Local vs Global Models in Robotics

🔹 Global Model (Deliberative)

- Represents the robot's **overall environment** and **mission objectives**.

- Used in **path planning, map-making, mission sequencing**, and decision-making.

- Based on **deliberative control**, often using world models (e.g., occupancy grids, topological maps).

- **Slow but informed** decisions.

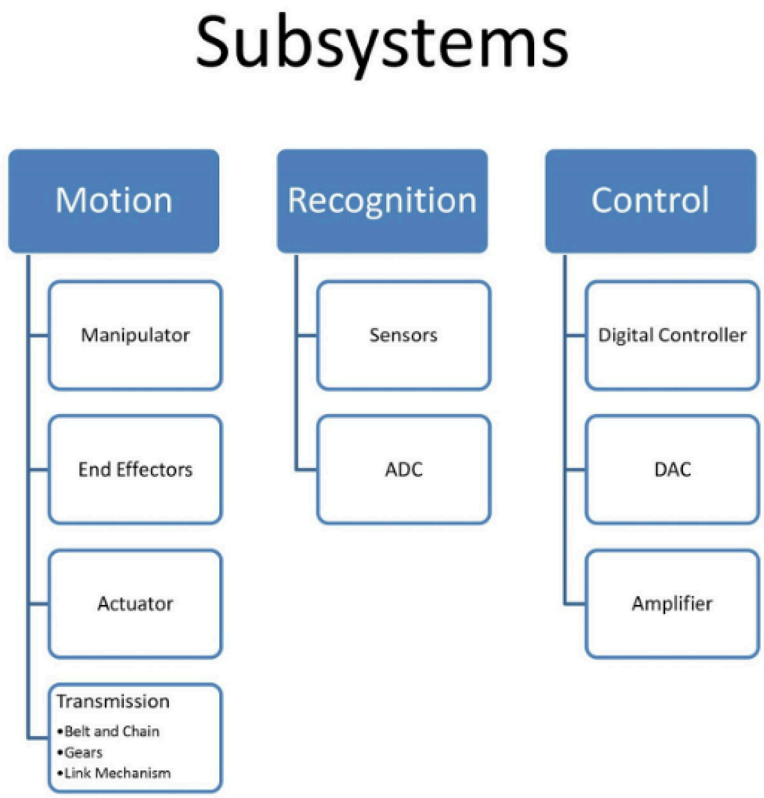
🔹 Local Model (Reactive)

- Immediate perception of the **local surroundings** (e.g., obstacle right now, current slope).

- Enables **quick, reflex-like behavior** such as collision avoidance or wall-following.

- Used for **reactive control** - fast responses without needing full knowledge.

However; Robot behavioral management requires to know its current mission, state and environment beside path-planning, map-making, monitoring etc. So, both local and global models are required to be considered for a robot performance.



This diagram shows the three main subsystems of a robot: **Motion**, **Recognition**, and **Control**. The **Motion subsystem** includes the manipulator, end effectors, actuators, and transmission systems (like gears or chains), all responsible for moving and interacting with the environment. The **Recognition subsystem** uses sensors to perceive the surroundings and ADCs to convert sensor signals into digital data. The **Control subsystem** consists of a digital controller that makes decisions, DACs that translate digital outputs into analog signals, and amplifiers that boost signals to drive actuators. Together, these subsystems enable a robot to sense, process, and act.

🔹 Motion Subsystem

Responsible for all **physical movement** of the robot.

- **Example:** In a robotic arm (like in car assembly lines):
  - The **manipulator** is the full arm structure with joints.
  - The **end effector** is the gripper or welding tool at the end.
  - **Actuators** (like servo motors) move the joints.
  - **Transmission** (gears or belts) delivers motor torque to the joints.

So, when the robot needs to pick up a part, the motion subsystem physically moves the arm to the right position.

🔹 **Recognition Subsystem**

Handles **perception and sensing** of the robot's environment.

- **Example:** In a line-following robot:
  - **Sensors** (IR or cameras) detect the path.
  - **ADC (Analog-to-Digital Converter)** turns the sensor's analog signal (voltage) into digital values the controller can process.

This allows the robot to "see" or detect what's around it and react accordingly.

🔹 **Control Subsystem**

Makes decisions and **sends signals** to the motion components.

- **Example:** In a drone:

- The **digital controller** (e.g., Arduino) calculates how to stabilize the drone or follow GPS coordinates.
- **DAC** converts digital control outputs into analog signals if motors need it.
- **Amplifiers** boost those signals to power the drone's motors.

This system ensures the drone flies smoothly by constantly adjusting motor speeds based on sensor feedback.

Example Workflow (Full Cycle):

A robot vacuum:

1. **Recognition**: Detects walls or dirt using IR or camera sensors.
2. **Control**: Calculates a new path to avoid obstacles.
3. **Motion**: Turns wheels using motors to move accordingly.

Each subsystem plays a role in making the robot autonomous and responsive.

🔹 Motion Subsystem

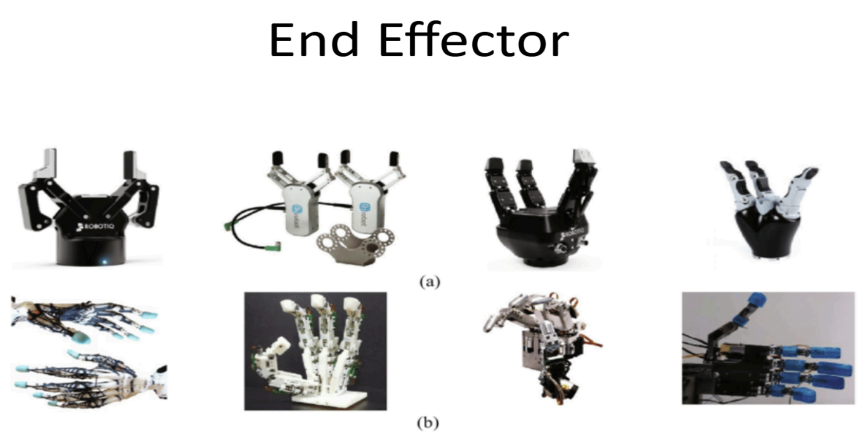
Manipulator

This likely depicts a **robotic manipulator**, which is a mechanical arm used to move objects with precision. Manipulators are the backbone of industrial robots.

It typically includes:

- **Links** (rigid segments),
- **Joints** (rotational or prismatic),
- **Base** (fixed or mobile),
- **End-effector** (tool or gripper at the end).

End Effector



Two types of fingered end-effectors: (a) gripper type, (b) anthropomorphic type

Shows two types of end-effectors:

1. Gripper Type – Simple mechanical fingers for pick-and-place.
2. Anthropomorphic Type – Mimics a human hand, more dexterous.

- Used for grasping, welding, painting, etc.

***Advantages of Chain, Belt, and Gearbox Transmission

Function	Gearbox	Chain Drive	Belt Drive
1. Motion Transfer	Very efficient, compact, minimal slip	Strong and reliable, minimal slip	Simple, quiet, but may slip under load
2. Speed Control	Precise control via gear ratios (fixed)	Speed varies with sprocket size ratio	Adjustable via pulley size; smooth transitions
3. Torque Control	Excellent - increases torque with high ratios	Good torque transmission, suitable for heavy loads	Moderate - can stretch or slip under high load
4. Direction Change	Possible with bevel or worm gears (90° etc.)	Usually fixed path, not great for directional change	Limited; requires complex pulley routing
5. Form Change	Rigid - only works with fixed alignment	Flexible routing, good for long-distance drives	Very flexible; can be used around obstacles

***Transmission systems:

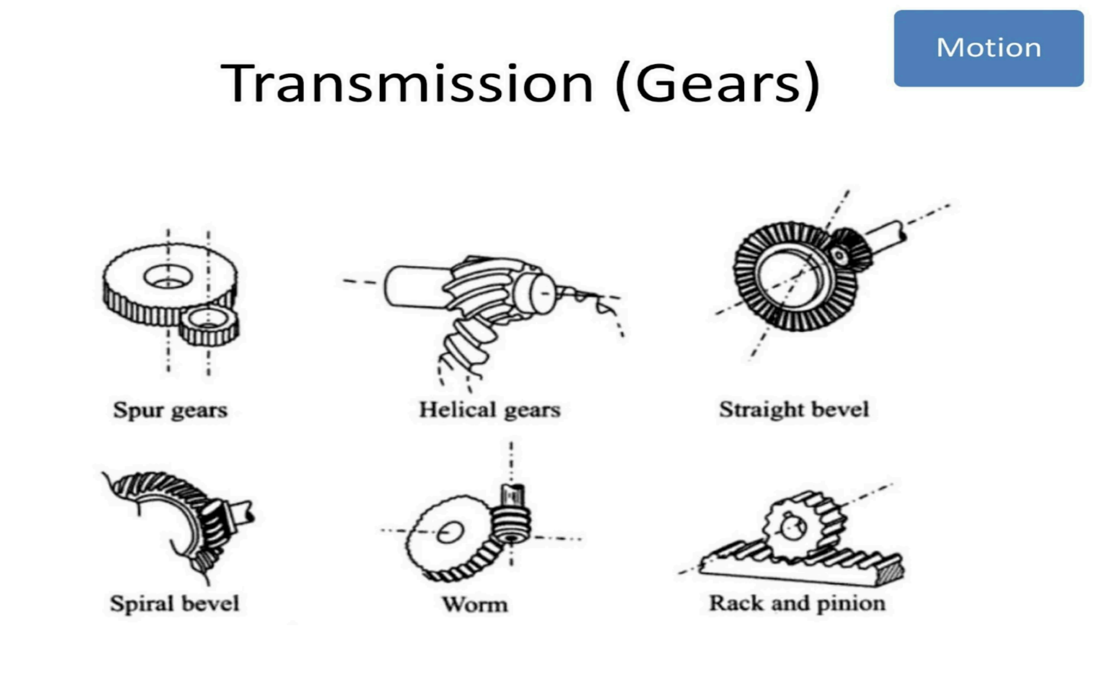
Transmission systems are essential components of robots and other machinery that convert rotational motion into linear or angular motion to achieve the desired output. Four common types of transmission systems used in robotics are belt transmission, chain transmission, gear transmission, and link mechanism transmission.

Belt Transmission: Belt transmission systems use flexible belts made of rubber or other materials to transfer rotational motion from one pulley to another. They are commonly used in applications of energy transmission, such as in conveyors, printing presses, and agricultural equipment. Belt transmission systems are simple, cost-effective, and easy to maintain, but they may slip under heavy loads or high speeds.

Chain Transmission: Chain transmission systems use roller chains to transfer power between two or more sprockets. They are commonly used in high-power, high-torque applications, such as in motorcycles, bicycles, and heavy machinery. Chain transmission systems are durable, long-lasting, and able to handle high loads and speeds. However, they may require regular lubrication and maintenance to prevent wear and tear.

Gear Transmission: Gear transmission systems use meshing gears to transmit rotational motion and power between two or more shafts. They are commonly used in applications that require magnification or reduction of speed. It also helps to attain precise speed and torque control, such as in automobiles, airplanes, and machine tools. Gear transmission systems are highly efficient, compact, and able to transmit large amounts of power.

In summary, the choice of transmission system will depend on the specific requirements of the application, including factors such as speed, torque, precision, and environmental conditions. Each type of transmission system has its own strengths and limitations, and it is important to choose the right type of transmission system for the application in order to achieve optimal performance and efficiency.



🔹 Spur Gear

Purpose	Description
Simple and Efficient Power Transmission	Transfers motion between **parallel shafts**.
Low-Speed Robotics Systems	Used in basic gearboxes for **clean, low-speed, low-noise** operation.
Cost-Effective and Easy to Manufacture	Found in educational robots, small mobile robots, and prototyping.

🔹 Helical Gear

Purpose	Description
Smooth and Quiet Operation	Teeth are angled, so engagement is gradual, reducing vibration.
High-Speed or High-Load Applications	Used in industrial arms, AGVs, and mobile robots requiring higher torque and quieter drive.
More Torque Transmission than Spur	Suitable where performance and durability matter.

🔹 Straight Bevel Gear

Usage	Description
Changes Direction (90°)	Connects **intersecting shafts** — usually at right angles.
Low-Speed and Low-Torque Needs	Found in corner joints of robot arms, camera mounts, sensor turrets.

Simple and Rigid Design	Cheaper alternative to spiral bevels for straightforward applications.

🔹 Spiral Bevel Gear

Usage	Description
Smooth Direction Change	Similar to straight bevel, but with **curved teeth** for smoother motion.
High-Speed & High-Torque Corners	Used in precision robot arms, humanoid hip/shoulder joints.
Less Noise, More Strength	Preferred over straight bevel when motion quality matters.

🔹 Worm Gear

Purpose	Description

High Torque Output	Ideal for lifting or rotating heavy loads with small motors.
Self-Locking Mechanism	Prevents back-driving; holds position without power.
Compact Gear Reduction	Achieves high gear ratios in small space, suitable for robot joints.
Direction Change (90°)	Transmits motion perpendicularly, useful in confined spaces.
Quiet Operation	Low noise; useful in medical or indoor service robots.

🔹 Rack and Pinion Gear

Purpose	Description
Convert Rotation to Linear Motion	Used where motors need to move components straight (e.g., sliders, lifters).

Precise Linear Positioning	Common in robot arms, 3D printers, and inspection platforms.
Heavy Load Handling	Rigid, strong design suited for gantry robots and vehicle platforms.
Steering Mechanism	Employed in steering systems of mobile robots or autonomous vehicles.

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1. DC Motor

Advantages:

● Simple and easy to use.
● Provides high starting torque.
● Low cost compared to other types of motors.
● Can operate over a wide range of temperatures and environments.

Limitations:

● Cannot maintain a constant speed under varying loads.
● May experience reduced efficiency at low speeds.
● Requires regular maintenance to ensure proper operation.
● No angular control.

Uses:

● DC motors are useful when you need a simple, low-cost motor that can provide high starting torque and speed control.
● They are commonly used in applications such as electric vehicles, industrial machinery, and robotics.

2. Stepper Motor

Advantages:

● Provides precise control over positioning and speed.

● Can maintain a constant speed under varying loads.

● Low cost compared to servo motors.

● Provides high torque output at low speeds.

● Easy to use and control.

Limitations:

● Less efficient than DC motors at high speeds.

● Limited speed range.

● Can experience increased electrical noise output.

Uses:

● Stepper motors are useful when you need precise positioning control, such as in CNC machines, 3D printers, and robotics.

● They can also be used in applications where low-speed torque is required, such as in surveillance cameras or robotics.

3. Servo Motor

Advantages:

● Provides accurate and precise control over position, velocity, and acceleration.

● Can maintain a constant speed under varying loads.

● Low electrical noise output.

Limitations:

● May require more complex control circuitry.

● Sometimes low.

● Can be sensitive to vibration and shock.

● Limited angle of rotation.

● Requires precise calibration to achieve optimal performance.

Uses:

● Servomotors are useful when you need accurate and precise control over the position, velocity, and acceleration of the motor.

● They are commonly used in applications such as robotics, industrial automation, and aerospace systems.

4. Induction Motor

- Runs on AC power.
- The rotor gets power by induction (no brushes).
- Simple, robust, cheap, good for heavy loads.
- Used in fans, pumps, and industrial machines.

Remember:

- DC = continuous spin with voltage control
- Stepper = move step-by-step with pulses
- Servo = precise position with feedback
- Induction = AC motor, strong and simple

***Problems of DC Motor

- Brushes wear out and need replacing.
- Sparking causes noise and reduces reliability.
- Limited maximum speed due to mechanical parts.
- Efficiency loss from friction and heat.
- Usually bigger and heavier than some other motors.

***Solutions to DC Motor Problems

- Use **brushless DC motors** to avoid brush wear and sparking.
- Regular **maintenance** to replace worn brushes on brushed motors.
- Use **better materials** for brushes to reduce wear and noise.
- Implement **cooling systems** to manage heat and improve efficiency.

- For high-speed needs, switch to **brushless or other motor types** like stepper or servo motors.

Stepper vs Servo Motor

- **Stepper motor:**
  - Moves in precise, fixed steps (open-loop control).
  - Cheaper and easier to use.
  - Best for low-speed, low-torque, simple positioning tasks.
- **Servo motor:**
  - Uses feedback for precise control of position, speed, and torque.
  - Handles high-speed, high-torque, and complex tasks.
  - More accurate and powerful than stepper motors.

In short: Stepper motors work well for simple, less demanding tasks, but servo motors are better suited for precise and heavy-duty applications.

***Actuators :

In robotics, an actuator is a component that is used to convert energy into motion or force. The actuator is responsible for moving the robotic system or its components, such as the manipulator or end effector, in a controlled manner. There are several different types of actuators that are commonly used in robotics, each with its own strengths and limitations.

Electric Actuators: Electric actuators are the most common type of actuators used in robotics. They convert electrical energy into motion or force, and can be controlled precisely using software or other control systems. Some examples of electric actuators include DC motors, stepper motors, and servo motors. Electric actuators are typically lightweight, compact, and easy to control, making them a popular choice for a wide range of applications. However, they may not be as powerful or able to generate as much force as hydraulic or pneumatic actuators.

Hydraulic Actuators: Hydraulic actuators use pressurized fluid, typically oil or water, to generate force and motion. They are often used in applications that require high levels of force or power, such as heavy-duty construction or industrial equipment. Hydraulic actuators can generate a lot of force and are able to maintain a constant force over long periods of time. However, they may be more complex and expensive to maintain than electric or pneumatic actuators, and may require a separate hydraulic system to operate.

Pneumatic Actuators: Pneumatic actuators use compressed air or other gasses to generate motion and force. They are often used in applications that require quick movements or high levels of precision, such as in assembly lines or robotics research. Pneumatic actuators are typically lightweight, easy to control, and able to generate force quickly.

Characteristics	Pneumatic	Hydraulic	Electric
Efficiency	Low	~~Low~~ High	High
Reliability	Excellent	Good	Good
Maintenance	High user-maintenance	High user-maintenance	Little to no maintenance
Purchase Cost	Low	High	~~High~~ Low
Operating Cost	Moderate	High	Low

🔹 Recognition Subsystem:

***Sensor :

A sensor is a device or subsystem that detects and responds to a physical stimulus, such as light, temperature, pressure, or motion, and converts it into an electrical or mechanical signal that can be processed, analyzed, or used to control other devices or systems. Sensors are essential components of many electronic, mechanical, and robotic systems, and are used in a wide range of applications, such as automotive, aerospace, medical, environmental monitoring, and consumer electronics. They can be classified based on the physical phenomenon they detect, the type of output they provide, and their operating principles. Some common types of sensors include temperature sensors, pressure sensors, proximity sensors, optical sensors, and motion sensors.

A sensor is a device that detects or measures physical, chemical or biological properties of the environment or a system and converts them into a signal that can be processed or analyzed.

Types of sensors:

Active and passive sensors are two different types of sensors used in various applications.

1. **Active sensors:** Active sensors emit energy or radiation to measure the properties of an object or environment. They require a power source to operate and can be used to gather data in situations where passive sensors are unable to function. Some examples of active sensors include <u>radar, lidar, and ultrasonic sensors.</u>

Radar sensors emit radio waves that are reflected back by objects in their path. By measuring the time it takes for the waves to return and the intensity of the reflection, radar sensors can determine the location, distance, and speed of objects.

Lidar sensors use laser beams to scan an area and create a 3D map of the environment. They measure the time it takes for the laser beams to reflect off of objects and return to the sensor, allowing them to create highly detailed and accurate maps.

Ultrasonic sensors emit high-frequency sound waves that bounce off objects and return to the sensor. By measuring the time it takes for the waves to return, ultrasonic sensors can determine the distance and location of objects. Application: measure wind speed and direction, navigation of UAV, measure tank depth.

2. **Passive sensors**: Passive sensors detect and measure the energy emitted by objects in the environment without emitting any energy themselves. They do not require a power source to operate and are often used in situations where active sensors may interfere with the environment or other sensors. Some examples of passive sensors include infrared sensors and optical sensors.

Infrared sensors is an electronic device that can detect and measure infrared (IR) radiation in the environment. Anything that emits heat (everything that has a temperature above around five degrees Kelvin) gives off infrared radiation. Application: TV Remote, Motion Sensing, Proximity Sensing.

Optical sensors use light to detect and measure various physical properties, such as color, intensity, and wavelength. They can be used in applications such as image sensing, position sensing, and spectroscopy.

LIDAR:

LIDAR (Light Detection and Ranging) is a remote sensing technology that uses laser light to measure the distance to objects in its field of view. LIDAR technology has many applications, including autonomous vehicles, surveying, mapping, and atmospheric monitoring.

LIDAR systems typically consist of a laser scanner, a receiver, and a data processing unit. The laser scanner emits laser pulses that travel through the atmosphere and bounce off objects in their path. The receiver detects the reflected laser pulses and measures their time of flight and intensity. The data processing unit then uses this information to create a 3D map of the environment.

There are two main categories of LIDAR systems, including:

1. **Airborne LIDAR**: These LIDAR systems are mounted on aircraft and are used for mapping large areas, such as forests, coastlines, and urban areas. Airborne LIDAR systems can cover large areas quickly and provide high-resolution maps.

2. **Terrestrial LIDAR**: These LIDAR systems are used for mapping smaller areas, such as building interiors, archaeological sites, and construction sites. Terrestrial LIDAR systems are mounted on tripods and can provide high-resolution 3D maps.

LIDAR technology has revolutionized many industries by providing accurate and detailed 3D maps of the environment. It has become an essential tool for autonomous vehicles, robotics, and remote sensing applications. However, LIDAR systems can be expensive and require skilled operators to operate and maintain.

🔹 Control Subsystem:

Control hardware in robotics refers to the controller that continuously reads from sensors like motor encoders, force sensors, or even vision or depth sensors, and updates the robot’s movements accordingly.

Microcontroller: Microcontrollers can interact with the physical world through sensors and actuators. Sensors are devices that detect and respond to changes in the environment, such as temperature, light, or motion. The microcontroller can read data from sensors and use it to make decisions or trigger actions. Actuators, on the other hand, are devices that can move or control a mechanism or system. The microcontroller can send signals to actuators to control their behavior.

For example, in a temperature control system, a temperature sensor might detect the current temperature and send that information to the microcontroller. The microcontroller can then compare the current temperature to a desired setpoint and send a signal to an actuator, such as a heater or fan, to adjust the temperature accordingly.

Arduino : Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs from various sources such as sensors or buttons and turn them into outputs such as activating a motor or turning on an LED.

Questions:

1. What is 4D?
2. 4 Laws of AI Robots?
3. Thumb rules on the decision a robot uses.
4. What are the 3 primitives of robotics?
5. What are the paradigms of robotics?
6. Differentiate between paradigm (Reactive vs Hybrid or Deliberative vs Hybrid)
7. What are the Subsystems?
8. Write ⅘ types of end effectors.
9. Write ⅘ names of sensors.
10. Write ⅘ types of gears.

Quiz Questions:

1.What are the main differences between Robotics and Automatic Machines? How Are Robotics and AI Shaping the 4th Industrial Revolution?

Main Differences Between Robotics and Automatic Machines:

Feature	Robotics	Automatic Machines
Definition	Machines that can sense, think (optionally with AI), and act	Machines that perform **predefined** tasks automatically
Flexibility	**Flexible** - can be reprogrammed to perform different tasks	**Fixed** - usually designed for one specific task
Sensors	Often includes **sensors** (vision, touch, etc.) to adapt to environment	Usually lacks adaptive sensors; uses timers or basic feedback loops
Intelligence	May use **AI** for decision-making	Follows a strict, **non-intelligent** control system
Examples	Robotic arm that adjusts to item positions	Washing machine, vending machine, old assembly line systems
Autonomy	Can function in **uncertain or changing environments**	Works well in **structured, fixed environments**

How Robotics and AI Are Shaping the 4th Industrial Revolution:

Robotics and AI are the driving forces of the **4th Industrial Revolution (Industry 4.0)**, transforming industries through **smart automation**, **real-time decision-making**, and **intelligent machines**.

Key Impacts:

1. **Smart Factories**
    - AI + robots enable automated, self-optimizing production lines.
2. **Flexible Automation**
    - Robots can adapt to different tasks; not just fixed routines.
3. **Human-Robot Collaboration**
    - Cobots work safely with humans, improving productivity and safety.
4. **Predictive Maintenance**
    - AI predicts machine failures, reducing downtime.
5. **Supply Chain Optimization**
    - AI and autonomous robots streamline logistics, warehousing, and delivery.
6. **Mass Customization**
    - Robotics and AI support personalized production without slowing down output.

2. Write-down your overall assessment on Robotics.

Robotics is a rapidly evolving field that is revolutionizing the way we live and work. It combines **mechanical engineering**, **electronics**, and **computer science** to create machines that can sense, think, and act in the physical world. From manufacturing and healthcare to agriculture and space exploration, robots are becoming essential tools in performing repetitive, dangerous, and complex tasks with **precision, speed, and consistency**.

As technology advances, robots are becoming **smarter** through integration with **Artificial Intelligence (AI)**, enabling them to adapt, learn, and interact more naturally with humans and their environment. This makes robotics a critical component of the **4th Industrial Revolution**, driving innovation, increasing efficiency, and opening new possibilities across industries.

However, robotics also raises important ethical, social, and economic questions, such as job displacement, privacy concerns, and the need for regulation and human oversight. Overall, robotics holds **tremendous potential** to improve quality of life and transform industries, but it must be developed and deployed **responsibly**.

3. In Bangladesh, the population is still high and people can be easily found for labor work. Should we adopt robots in the garment sector? Why explain? Does it violate any law on the use of robots?

Should we adopt robots in the garments sector of Bangladesh?

Yes, but **gradually and strategically**.
While Bangladesh has a large labor force, adopting robots can **increase productivity, improve product quality, and meet global demands** faster. Robots can be used to perform **repetitive or hazardous tasks**, making the work environment safer. However, full automation is not ideal in a country where garments provide employment to millions. A **hybrid approach**, where robots assist rather than replace workers, would be more appropriate.

Why should we adopt them?

- To improve **production speed and accuracy**
- To maintain **global competitiveness** with other countries using automation
- To reduce **workplace injuries** and **physical strain** on workers
- To handle **large and urgent orders** efficiently
- To support **quality control** and **consistency**

Does it violate any law of robot use?

It may **violate ethical and social responsibilities** if automation causes mass **job loss without support** for displaced workers. While not illegal, replacing human workers without planning for reskilling and alternative employment can be considered a **violation of the fair use principle** of robotics, which promotes **human welfare, not harm**.

4.What are common problems with DC motors, and how can they be solved?

5. Explain them. i) Gear, ii) Servo motor, iii) raspberry pi, iv) hydraulic

6.Bracu Duburi is participating in a RoboSub- an Underwater Robotic Competition. What kind of robot is this? Write down its characteristics.

RoboSub is an underwater robot called an Autonomous Underwater Vehicle (AUV).

Characteristics:

- Works autonomously without human control

- Waterproof and pressure-resistant
- Moves using thrusters in all directions
- Uses sensors like sonar and cameras to navigate
- Performs tasks like object detection and retrieval underwater
- Battery-powered with limited communication

7.Is the hybrid paradigm more effective than the deliberative and reactive paradigms?

The **hybrid paradigm** is generally better than purely **deliberative** or **reactive** paradigms because it combines the strengths of both. It offers the **goal-directed planning** of the deliberative approach and the **real-time adaptability** of the reactive approach. This makes it more suitable for complex and dynamic environments like robotics and autonomous systems.

Real-Life Example:

- **Self-driving cars**:
  - Use **deliberative** for route planning.
  - Use **reactive** for obstacle avoidance.
  - So they follow a **hybrid approach** to ensure both **safety** and **goal achievement.**